

Distance Metric Geometry for Regime Detection in Synchrophasor Data

Comparing Classical and Quantum-Inspired Approaches

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Why Monitor Grid Oscillations?

THE PROBLEM

Power grids operate near stability limits. Oscillations can cascade into blackouts if not detected early. High-dimensional signal data at petabyte scale across thousands of streams requires a lot of processing power

TRADITIONAL METHODS STRUGGLE

Model-based approaches require accurate system models. Signal processing methods need manual tuning. Scale is always a bottleneck

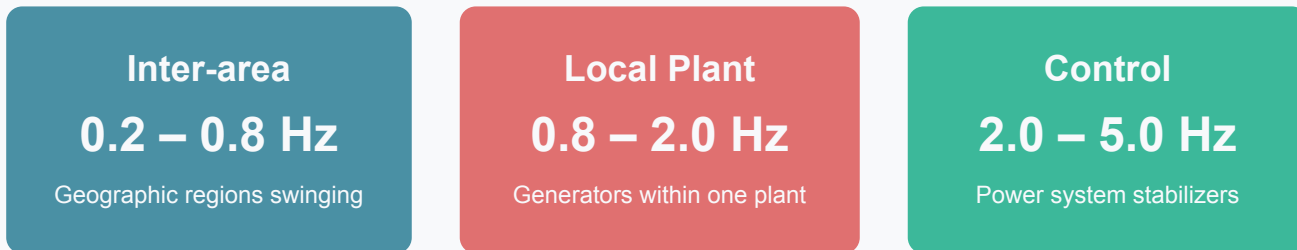
THE APPROACH

Two families of distance metrics (each with classical vs quantum variants) compared between sequential PSD vectors for segmented signal with TDA for mode detection per segment

0.2 – 5 Hz

Oscillation frequency range of interest

Oscillation Mode Categories



Each mode type requires different detection sensitivities

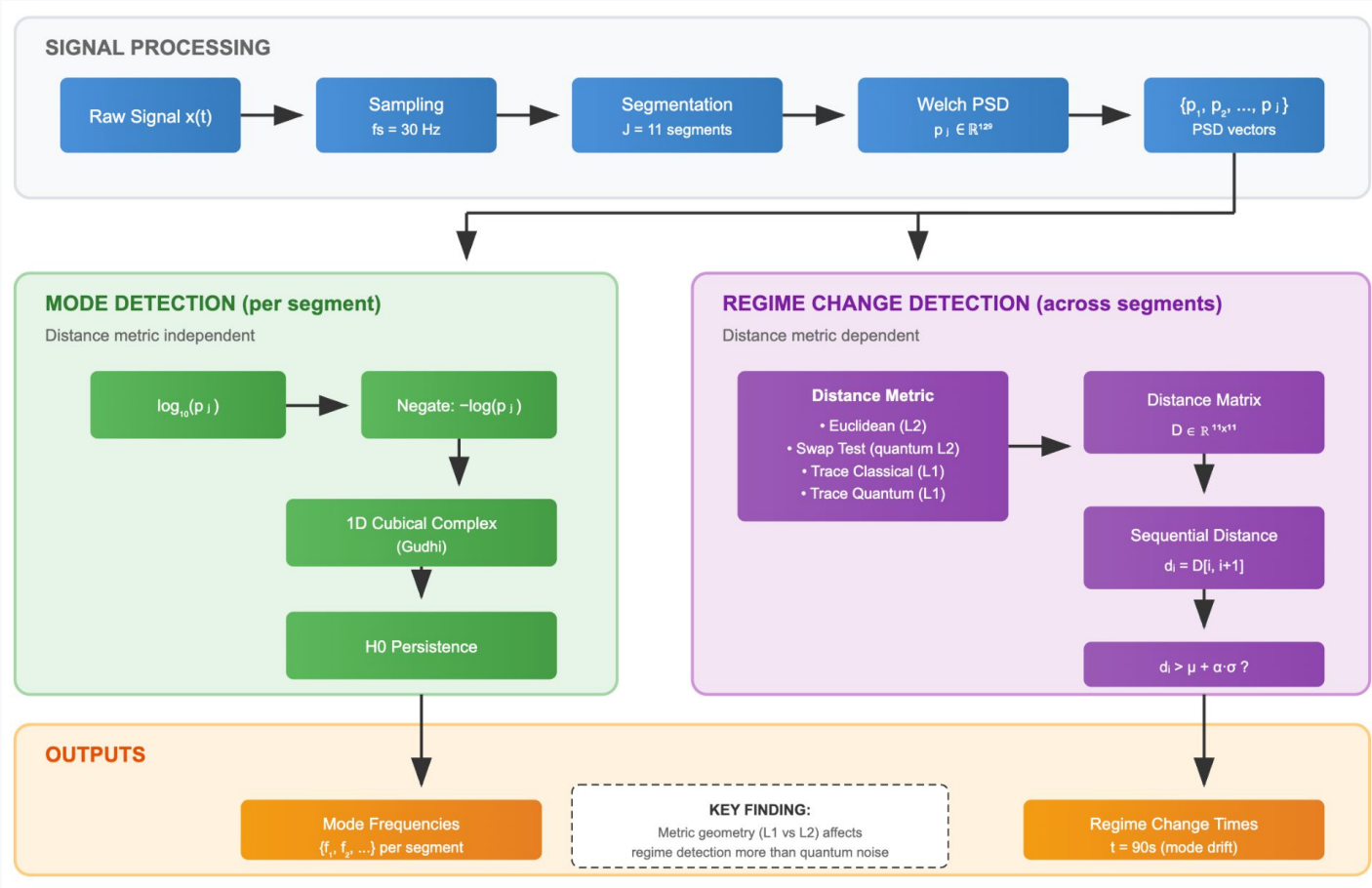
GESL signals in our study: primarily inter-area (0.55, 0.75 Hz)

Hypothesis

Metric geometry (L1 vs L2)

matters more than
quantum vs classical

Analysis Pipeline



Four Distance Methods Compared

L2 FAMILY (Euclidean)

Classical Euclidean

Standard L2 norm between raw PSD vectors

Range: $[0, \infty)$ unbounded

Quantum Swap Test

Inner product via quantum circuit, derive L2

+ sampling noise from finite shots

Responds to **magnitude** changes

L1 FAMILY (Trace)

Classical Trace Distance

Normalize PSD $\rightarrow$ probability, compute total variation

Range: $[0, 1]$ bounded

Quantum Sampled Trace Distance

Amplitude encoding, sample distribution

+ sampling noise from finite shots

Responds to **redistribution**

What Each Metric Detects

L2 (Euclidean)

"Did it get loud?"

Amplitude excitation events

Sudden power bursts

Sensitive to amplitude noise

Best for: intensity changes

L1 (Trace)

"Did the shape change?"

Frequency redistribution

Mode onset detection

May miss pure amplitude events

Best for: spectral shape

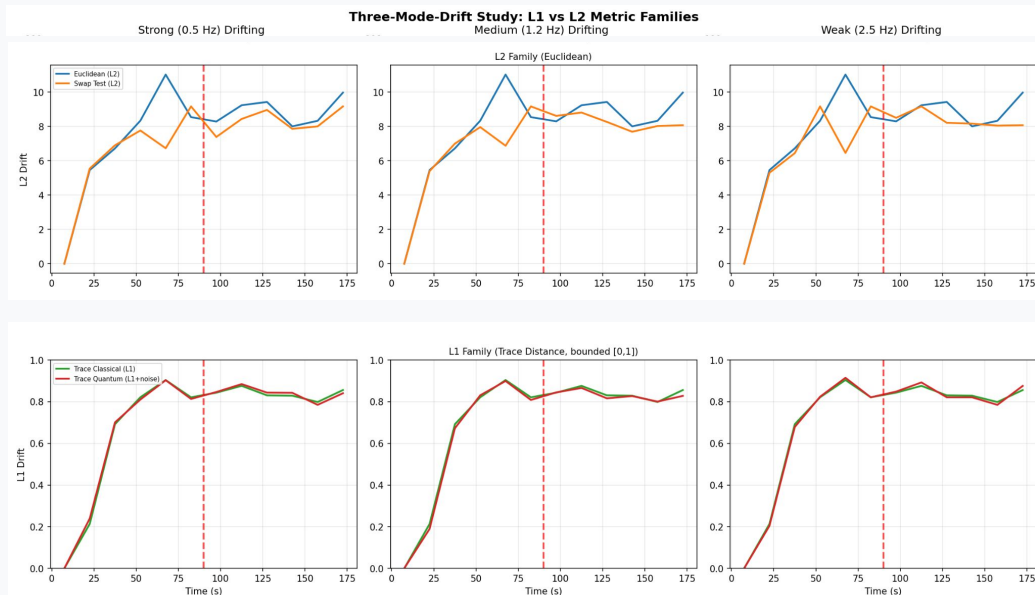
Simulated Data: Controlled Mode Drift

SETUP

- 3 modes (0.5, 1.2, 2.5 Hz)
- drift at $t=90s$
- 20 trials
- 180s duration

SCENARIOS

- Strong (0.5 Hz)
- Medium (1.2 Hz)
- Weak (2.5 Hz)



Method	Strong	Medium	Weak
Euclidean (L2)	72.5%	72.5%	72.5%
Swap Test (L2)	64.6%	67.9%	61.7%
Trace Classical (L1)	62.5%	62.5%	62.5%
Trace Quantum (L1)	65.0%	63.8%	64.6%

Real Data: Grid Event Signature Library

6 real PMU signals with known oscillation frequencies from PNNL's GESL repository



Metadata

Download

Filter Criteria

Display

Signature Id(s)

Description Contains

Event Date Range (UTC)

Data Sources

Uncheck All - 1 / 13 (Selected/Total)

☐ Provider 6

☐ Provider 7

☐ Provider 8

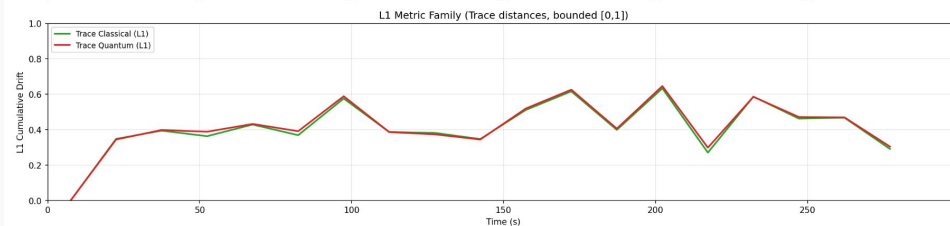
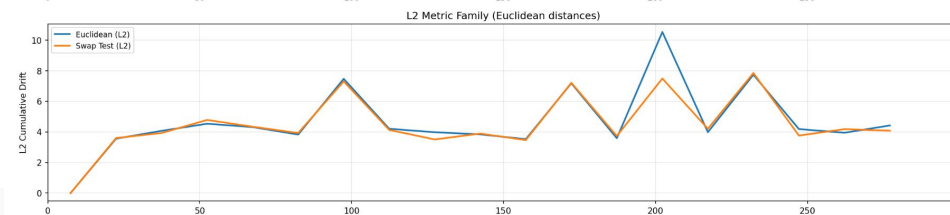
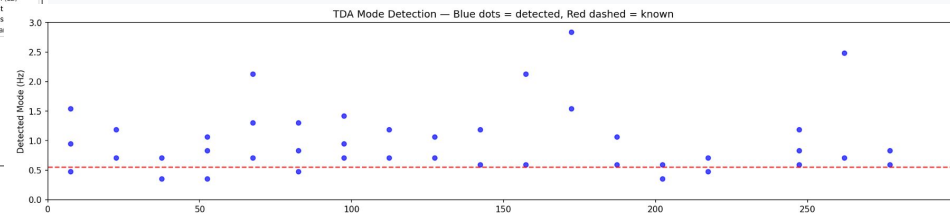
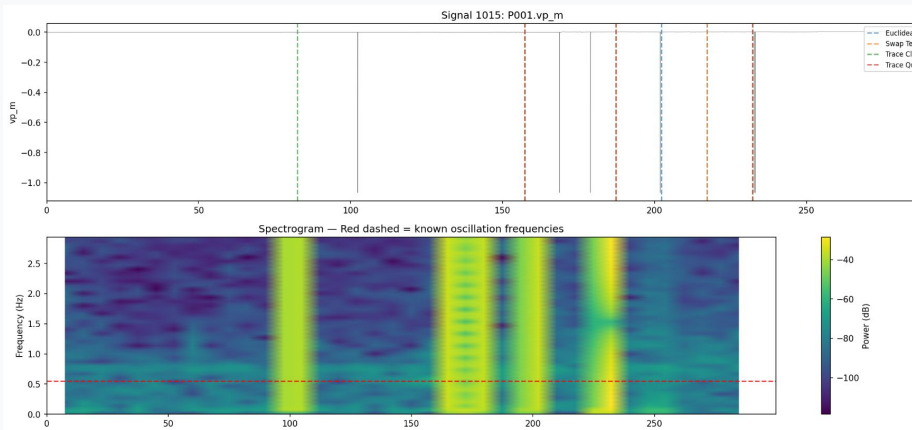
☒ Provider 9 (Eastern Interconnection)

☐ Provider 10 (Western Interconnection)

Total: **244 segments** analyzed across all signals

Real Data: Signal 1015

6 real PMU signals with known oscillation frequencies from PNNL's GESL repository

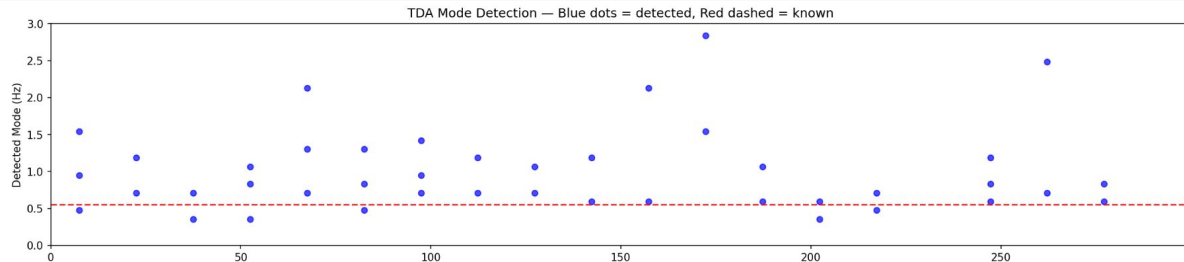


Total: **244 segments** analyzed across all signals

TDA Mode Detection: Signal 1015

Mean error: 0.042 Hz

Across all signals: mean error 0.036 Hz



Mode detection by segment: the 0.592 Hz (the 0.55Hz match) shows up at:

- 142s
- 157s
- 187s
- 202s
- 247s
- 277s

H0 persistence successfully identifies oscillation modes

Model-free detection

No manual tuning required

Works on raw PMU data

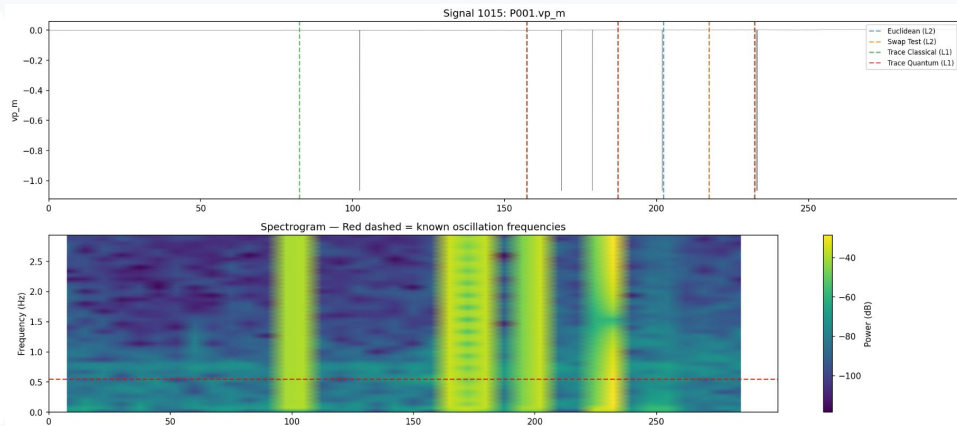
Note: 0.1 Hz modes in other signals at resolution floor show high error

L1 vs L2 Classical on Signal 1015

Four distinct events visible in spectrogram: **~around 100s**, **~175-180s**, **~200-210s**, and **~230-240s** with the strongest activity around **187-232s** where everyone agrees something happened

Regime changes detected:

- L1 (Trace Classical): 82s, 157s, 187s, 232s
- L2 (Euclidean): 187s, 202s, 217s, 232s

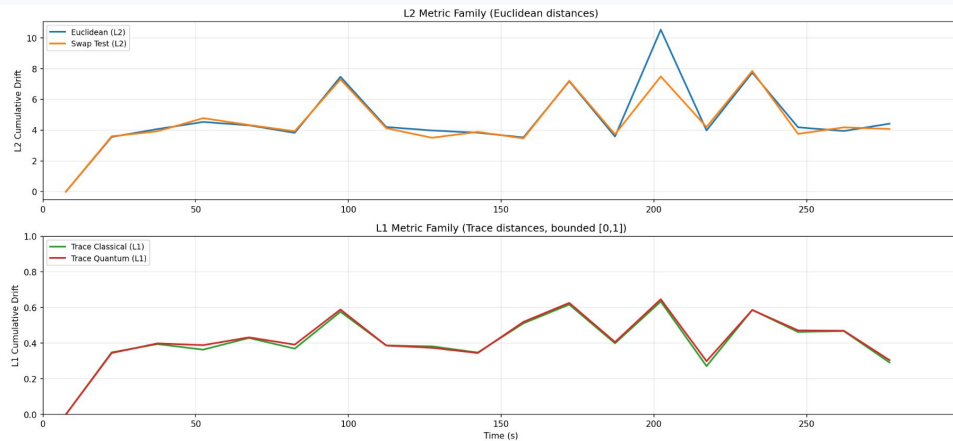


L1 excels when:

Mode onset is gradual, sustained structural changes, frequency redistribution

L2 excels when:

Amplitude bursts are the signal, ringdown events, intensity changes



L1 vs L2 Quantum-inspired on Signal 1015

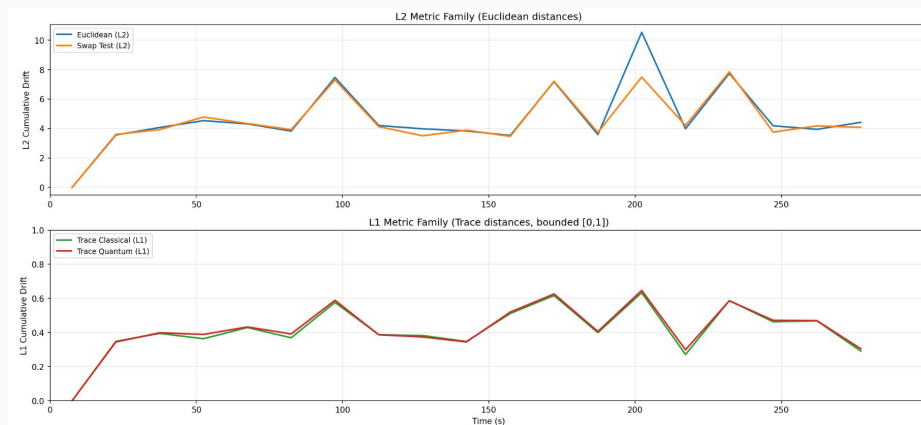
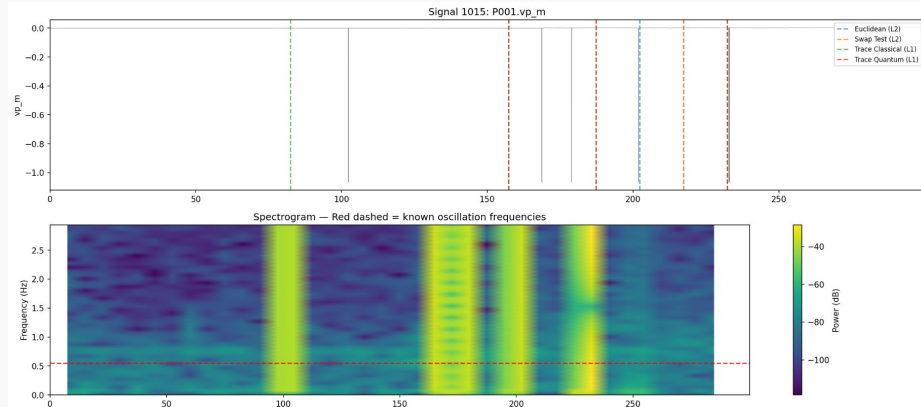
Swap Test: L2 via quantum circuit + shot noise.

Trace Quantum: Classical L1 with amplitude-encoded resampling.

Same underlying distances.

Same detection counts (± 1).

No obvious benefit, but a baseline for future experiments.



L1:

Classical: 4 detections

Quantum: 3 detections **Missed early warning at 82s** - noise cost a detection. Curves nearly identical - no new information, just jitter.

L2:

Euclidean: 4 detections

Swap Test: 4 detections **Caught 157s, missed 202s** - noise shuffles which events cross threshold. Cumulative drift tracks classical closely.

L1 vs L2 Quantum-Inspired on All Signals

NO ACCURACY BENEFIT

Quantum methods compute the same distances as classical, with added sampling noise from finite shots.

TIMING NOT MEANINGFUL

Used Qiskit simulator — overhead reflects classical emulation, not QPU performance.

VALUE OF THIS STUDY

Establishes accuracy baseline for future quantum algorithm development as hardware matures. Extensible pipeline for synchrophasor data analysis with modular elements.

Regime changes detected

Classical Euclidean	Swap Test	Trace Classical	Trace Quantum
40	41	41	41

Across all 6 GESL signals

Conclusions

1

Metric geometry dominates

L1 vs L2 matters more than quantum
vs classical

2

Complementary detection

L2 $\rightarrow$ amplitude, L1 $\rightarrow$ redistribution

3

TDA validated

0.04 Hz mean error on known modes

4

Quantum: noise, no benefit, a baseline

Sampling variance slightly degrades accuracy

5

Practical recommendation

Use both L1 and L2 in parallel

The Punchline

GRID OPERATORS

L1 catches what L2 misses.

Run both.

Minimal added cost.

QC COMMUNITY

Quantum computing $\neq$ quantum advantage.

This study is a baseline to beat.

SYNCHROPHASOR COMMUNITY

TDA works (0.036 Hz mean error)

No model.

No tuning.

Just data.

Practical, validated approach and implemented pipeline applicable today - quantum distance metric baseline for tomorrow

Next Steps

NEAR-TERM

Validate L1 on frequency-drift events. Compare against established mode drift algorithms. Extend to multi-PMU. Implement on QPU.

MEDIUM-TERM

Higher-dimensional TDA (H1), potentially incorporating quantum elements within TDA itself. Implement on point-on-wave signal (much finer resolution)

LONG-TERM

Large-scale grid analysis (1000+ PMUs). Comprehensive computational cost comparison

Key Open Question

Does L1's theoretical advantage for frequency migration hold on real mode-drift events?

(All GESL signals for this project showed amplitude excitation)

Thank You



Questions?

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